Distributed Systems

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Lecture 4

International Institute of Information Technology

Hyderabad, India

Vector Time

- One of the limitations of scalar time is the lack of strong consistency.
- Lack of strong consistency hinders tasks such as serialization.
- Strong consistency not achieved as the data structure used in scalar time – a single time – that represents the logical local clock is reused to represent the logical global clock.
  - This means that the causality of events across processors is lost.
- Vector time solves this problem but with big data structures.

Vector Time

- Time is represented by an n-dimensional non-negative integer vector.
- Each process p_i maintains a vector vt_i [1..n], where $vt_i[i]$ is the local logical clock of p_i and describes the logical time progress at process p_i.
- $Vt_i[j]$ represents process p_i 's latest knowledge of process p_j local time.
- If vt_i [j]=x, then process p_i knows that local time at process p_j has progressed till x.
- The entire vector vt_i constitutes p_i 's view of the global logical time and is used to timestamp events.

Vector Time – Update Rules

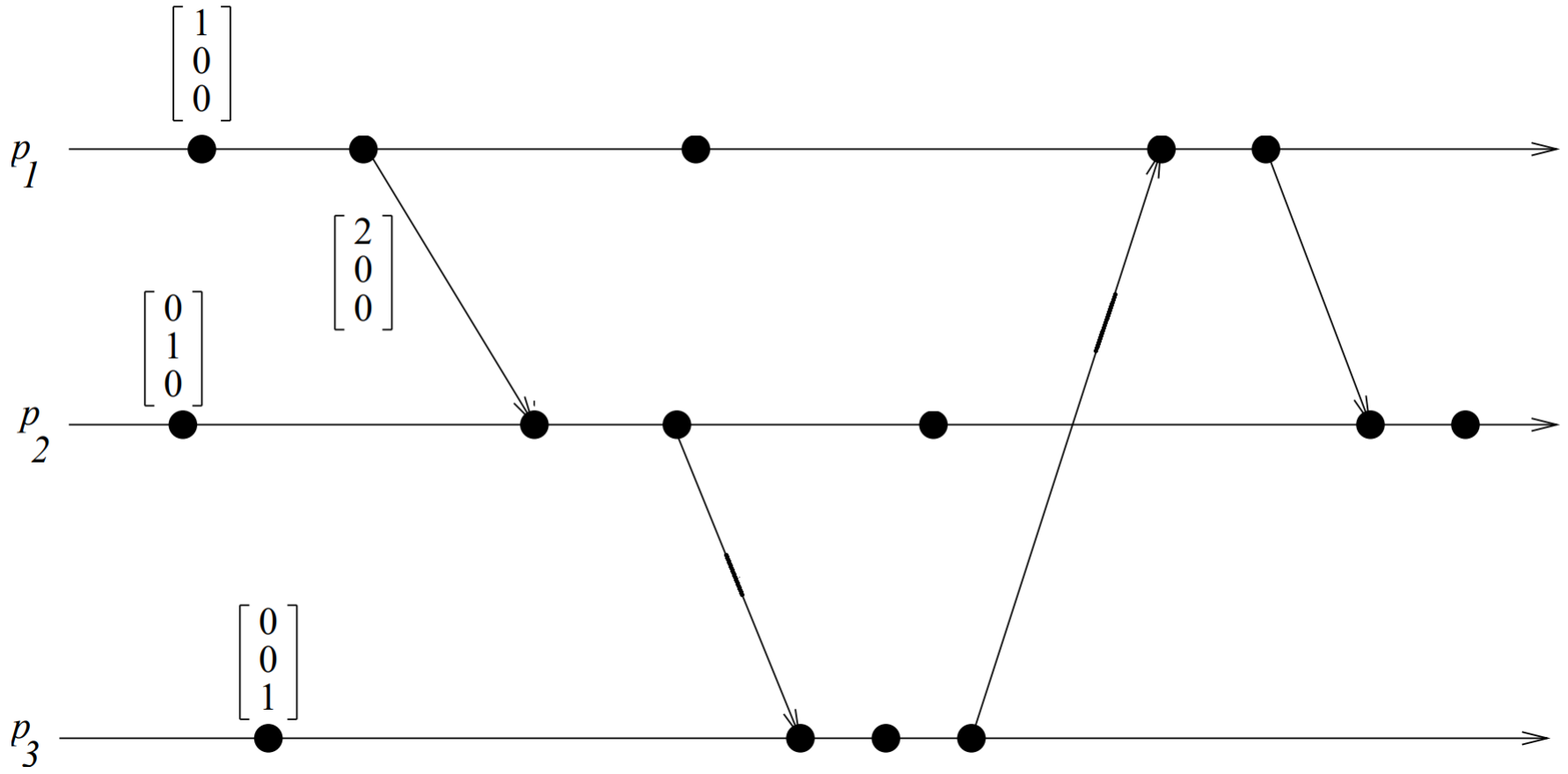
- Process p_i uses the following two rules to update its clock:
- Rule 1: Before executing an event, process p_i updates its local logical time as follows:

$$vt_i[i] := vt_i[i] + d \quad \text{where } (d > 0)$$

- Rule2: Each message m is piggybacked with the vector clock vt of the sender process at sending time. On the receipt of such a message (m,vt), process p_i executes the following sequence of actions:
  1. Update its global logical time as follows:
  2. $1 \le k \le n : vt_i[k] := \max(vt_i[k], vt[k])$
  3. Execute Rule1.
  4. Deliver the message m.
- The timestamp of an event is the value of the vector clock of its process when the event is executed.

Vector Time – Example

- Initially, the vector clock is set to [0, 0, 0, ..., 0] and d = 1.
- Fill the timestamps for the other events.



Vector Time – Example

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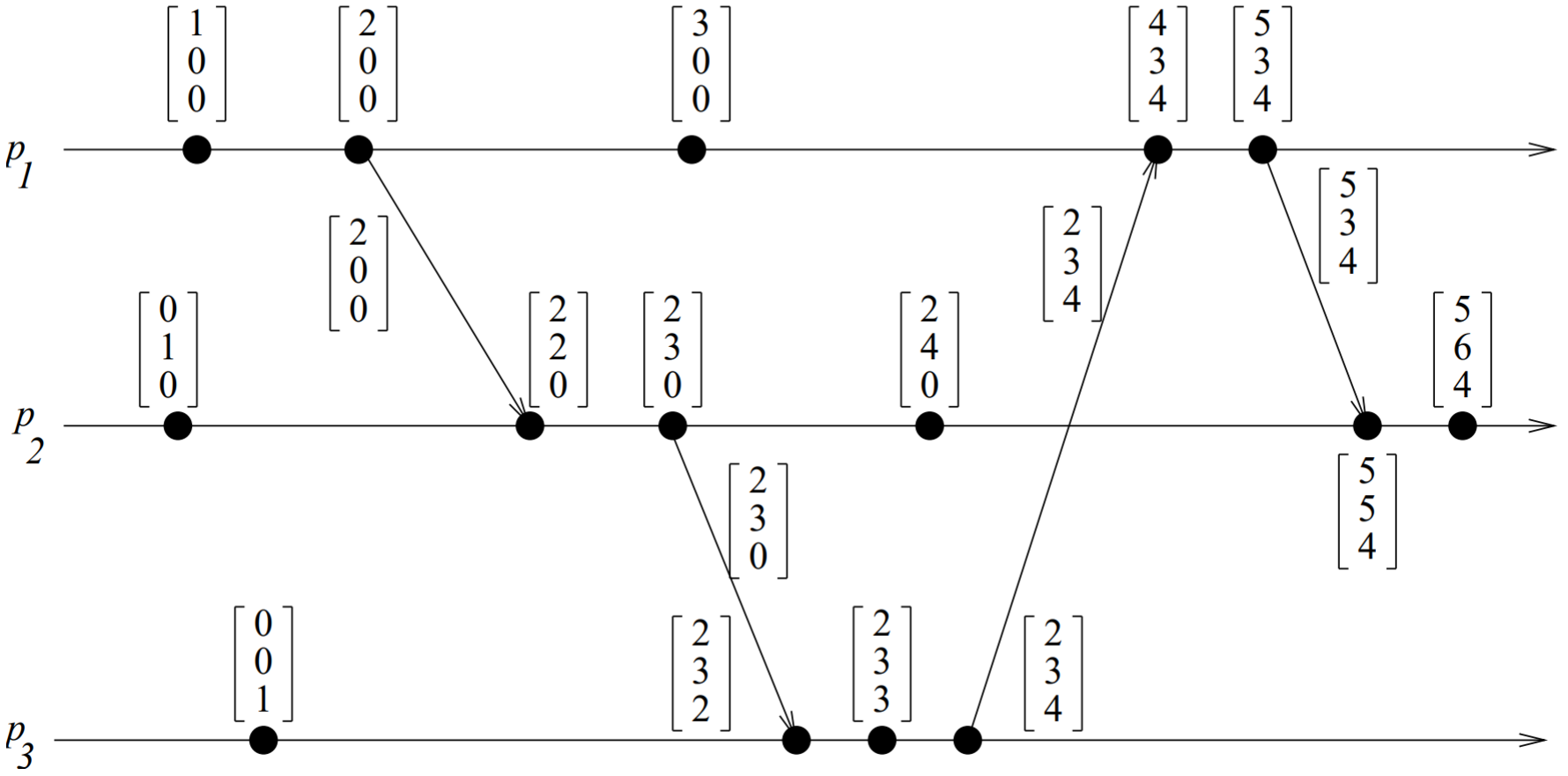


Vector Time

- Using vector clocks, two timestamps vh and vk are compared as follows.
  - vh == vk iff for all indices i, vh[i] == vk[i]
  - vh $\leq$ vk iff for all indices i, vh[i] $\leq$ vk[i]
  - vh < vk iff vh $\leq$ vk and there exists an index i where vh[i] < vk[i].
  - vh || vk iff not(vh < vk) and not(vk < vh)

- So, the vector [1, 3, 4] is less than the vector [1, 5, 6].
- The vectors [2, 5, 3] and [3, 4, 4] are concurrent.

Properties of Vector Time

- Event Counting: Use d = 1 always.
- Then the i th component of vector clock at process p_i,$vt_i[i]$, denotes the number of events that have occurred at p_i until that instant.
- So, if an event e has timestamp vh, vh[j] denotes the number of events executed by process p_j that causally precede e.
- Further, $\left(\Sigma_j \text{ vh[j]}\right) - 1$ represents the total number of events that causally precede e in the distributed computation.

Properties of Vector Time

- Strong Consistency: Vector clocks are strongly consistent.
- For any two events, we can determine if the events are causally related.
- Proof: Exercise.

Limitations of Vector Time

- Large message sizes owing to the vector being piggybacked on each message.
- The message overhead grows linearly with the number of processors in the system.
- When there are thousands of processors in the system, the message size becomes huge even if there are only a few events occurring in few processors.
- Few techniques exist to reduce the overhead.
  - Reading exercise.

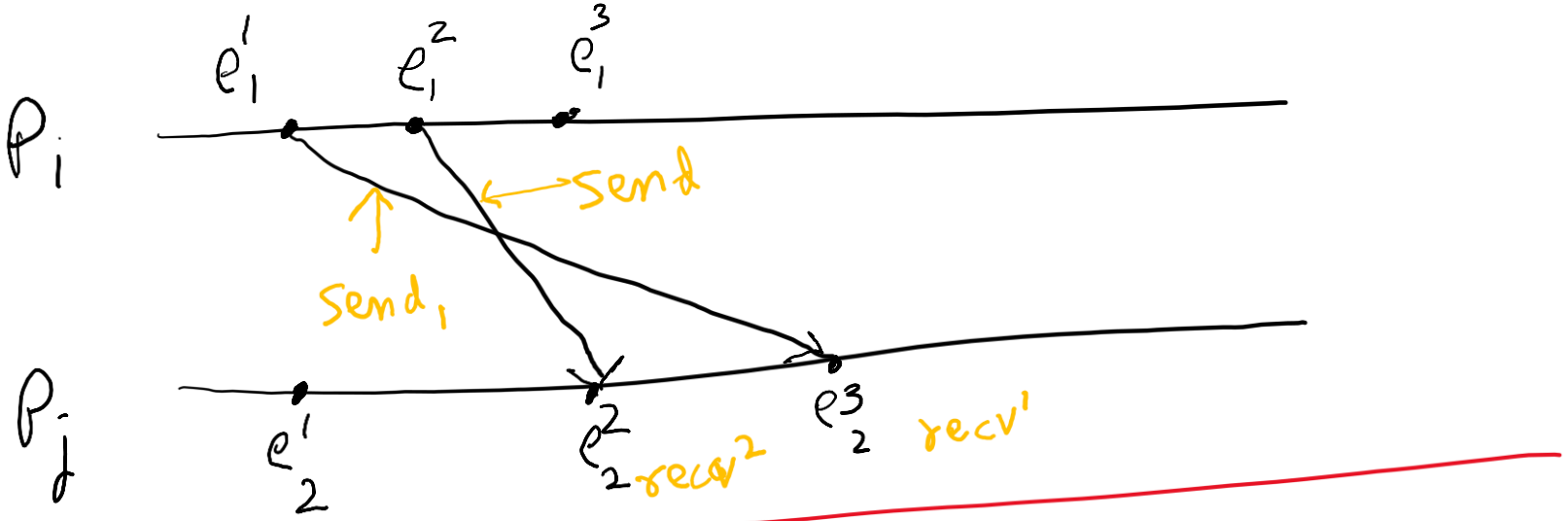
A Distributed Program in Execution

- Let us now model message delivery.
- Several ways are possible
  - Arbitrary : Message can be reordered in transit in the channel
  - FIFO : Messages are NOT reordered in transit in the channel
  - Causal : Slightly stronger than FIFO. Messages destined for the same destination are not reordered. In symbols,

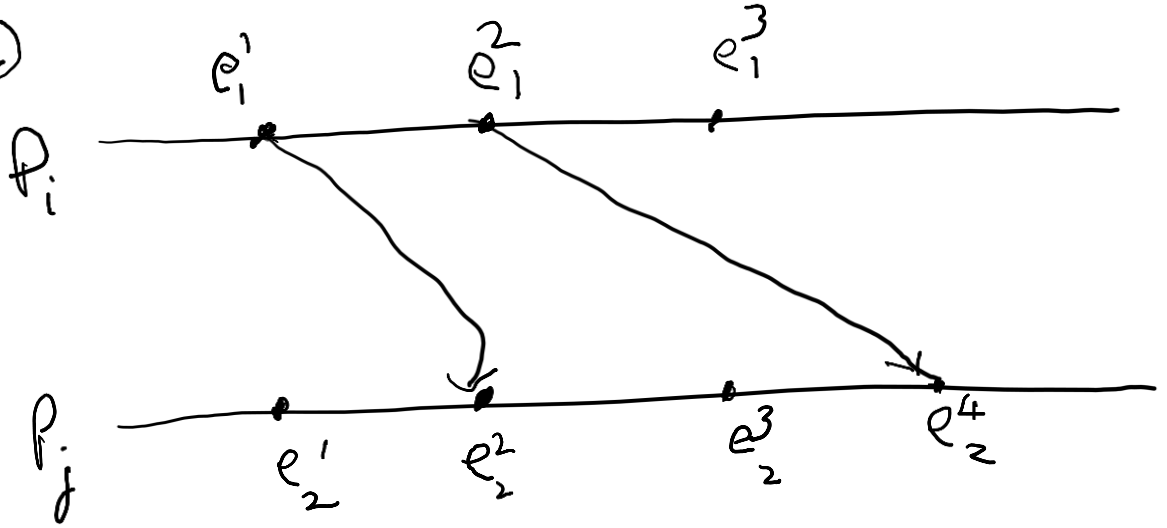
    for any two messages m_{ij} and m_{kj},
    if send(m_{ij}) $\rightarrow$ send(m_{kj}) then recv(m_{ij}) $\rightarrow$ recv(m_{kj}).

  - Causal $\subset$ FIFO $\subset$ Arbitrary
  - Draw pictures to understand the differences.

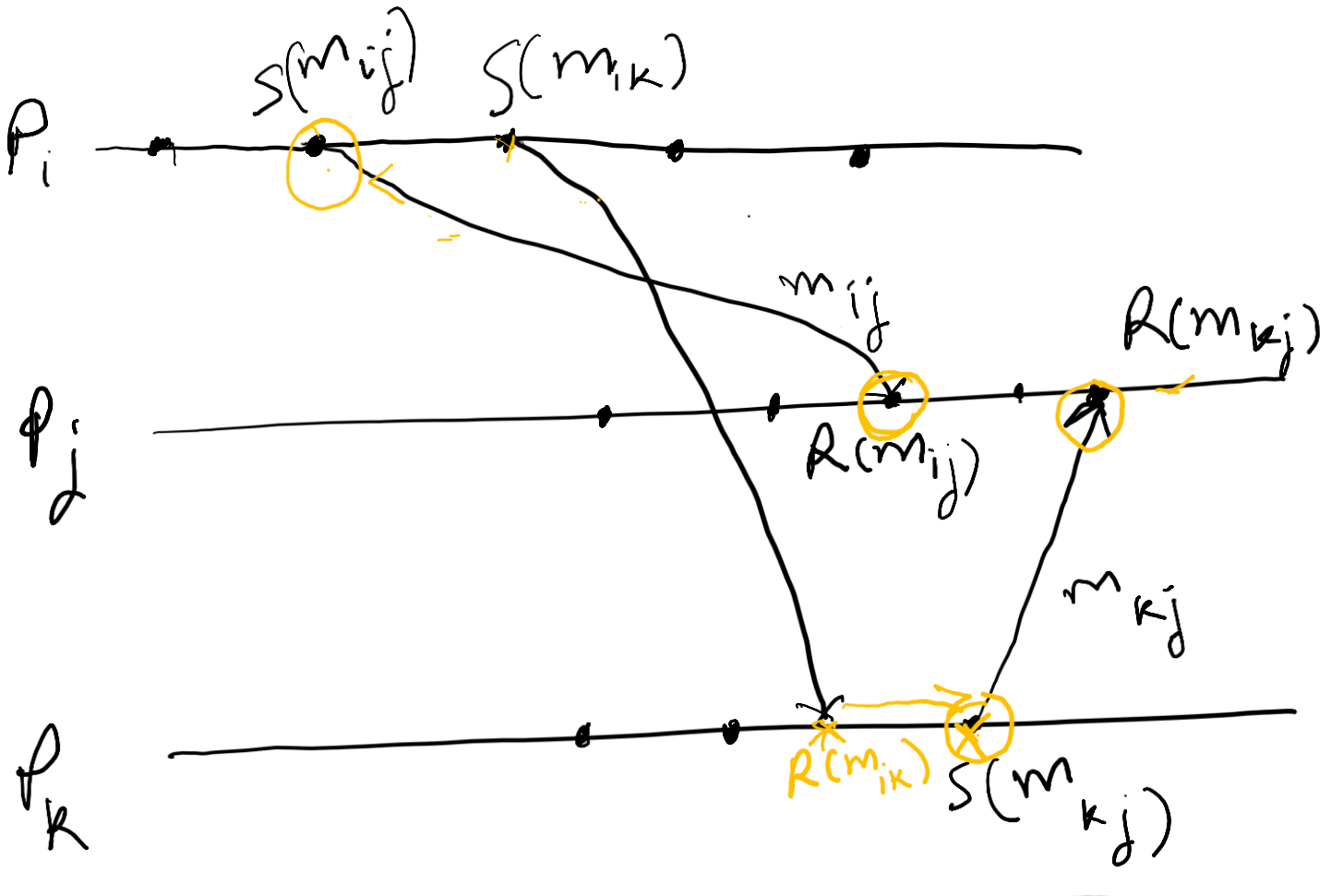
- Arbitrary



- FIFO



Causal



S send
R Recv.

$S(m_{ij}) \leq S(m_{kj})$

$\Downarrow$

$R(m_{ij}) \leq R(m_{kj})$

A Distributed Program in Execution

- Several ways are possible
  - Non-FIFO : Message can be reordered in transit
  - FIFO : Messages are NOT reordered in transit.
  - Causal : Slightly stronger than FIFO. Messages destined for the same destination are not reordered. In symbols,

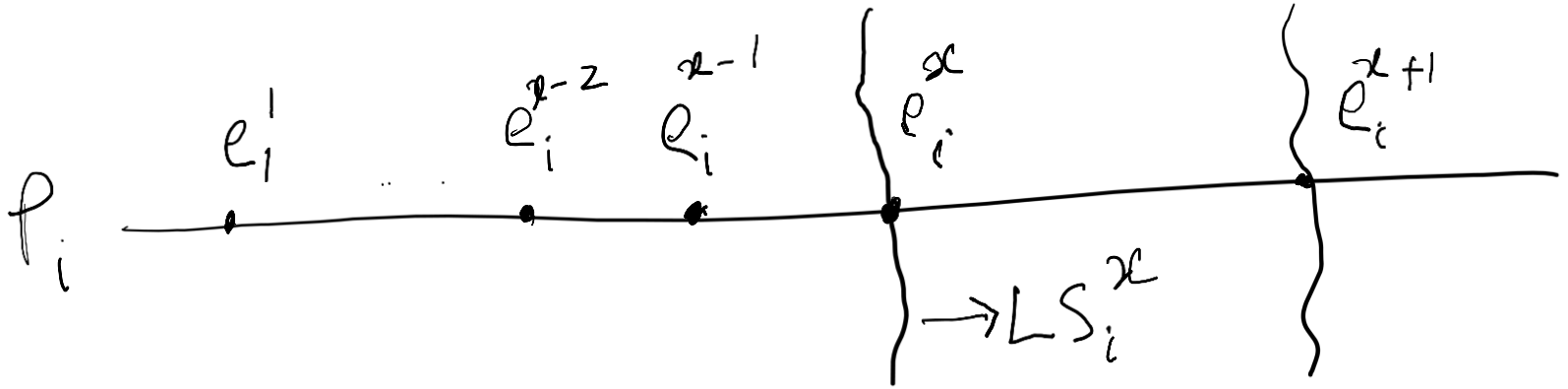
    for any two messages m_{ij} and m_{kj},
    if send(m_{ij}) $\rightarrow$ send(m_{kj}) then recv(m_{ij}) $\rightarrow$ recv(m_{kj}).

  - Causal $\subset$ FIFO $\subset$ Non-FIFO.
  - How can causal order among messages be guaranteed? Read about it from An Efficient Causal Order Algorithm for Message Delivery in Distributed System, Jangt, Park, Cho, and Yoon

The Global State of a Distributed System

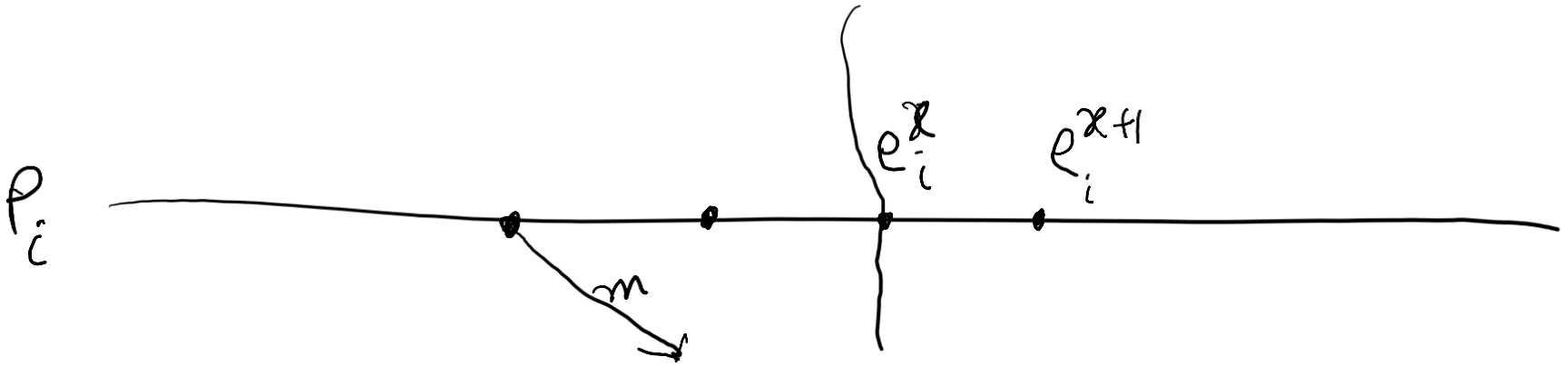
- One can think of the collection of local states as the global state.
- Recall that the local state of a process, P_i, is the contents of the processor registers, stack, local memory, etc.
- The state of a channel is the set of messages in transit in the channel.
- Events lead to changes in the state of local process(es).

The Global State of a Distributed System



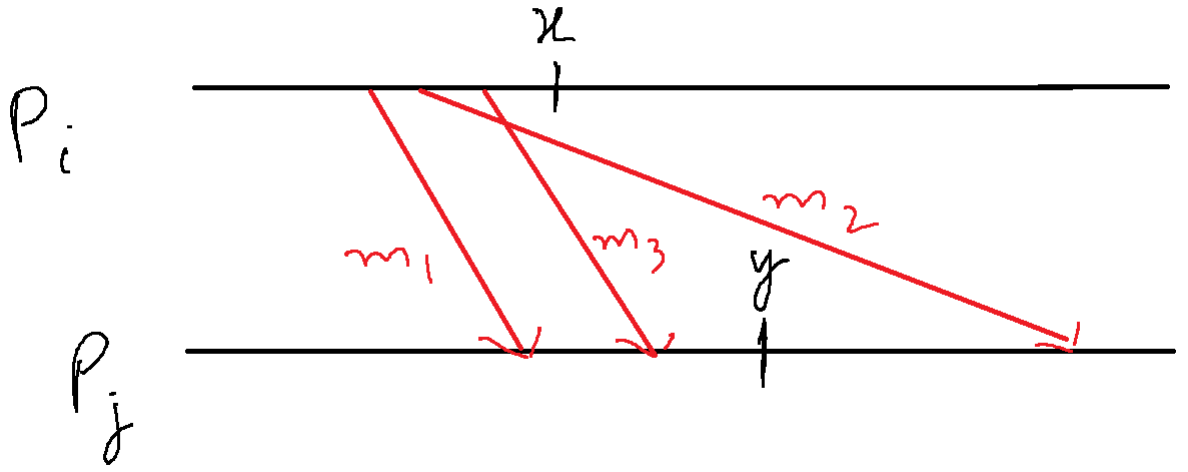
- To formalize, let LS^x_i denote the local state of process P_i after the occurrence of the event e^x_i and before the occurrence of the event e^{x+1}_i.
  - LS^0_i is the initial state of P_i
- By definition of LS^x_i, the effect of all the events before the event e^x_i are accounted for.

The Global State of a Distributed System



- For a message m and the event send(m), let send(m) $\leq$ LS^x_i denote that the message m is sent by process P_i on or before the event e^x_i.
  - There exists y s.t. $1 \leq y \leq x$, e^y_i= send(m).
- Likewise, can define recv(m) $\leq$ LS^x_i as
  - There exists y s.t. $1 \leq y \leq x$, e^y_i= recv(m).
- Can negate the above statements to indicate that send(m) not$\leq$ LS^x_i .

Global State of a Distributed System



- We use the above to capture the state of a channel C_{ij}, denoted $SC^{x,y}_{i,j}$ as follows.

$SC^{x,y}_{i,j} = \{m_{ij} \mid send(m_{ij}) \leq LS^x_i$ and $recv(m_{ij})$ not$\leq LS^y_j\}$

- In words, the state of $SC^{x,y}_{i,j}$ denotes all messages that have been sent by P_i up to event e^x_i and not received by P_j up to event e^y_j.

Global State of a Distributed System

- Finally, the global state of a distributed system, denoted GS, is defined as

$$GS = \{ \bigcup_i LS^{x_i}_i, \bigcup_{j,k} SC^{y_j z_k}_{jk} \}$$

- A global state is said to be consistent iff it satisfies the condition that a message m that is recorded as received is also recorded as sent in the state.

- Question: Write in symbols the above condition for a consistent global state.

Global State of a Distributed System

- Finally, the global state of a distributed system, denoted GS, is defined as

$$GS = \{ \bigcup_i LS^{x_i}_i, \bigcup_{j,k} SC^{y_j z_k}_{jk} \}$$

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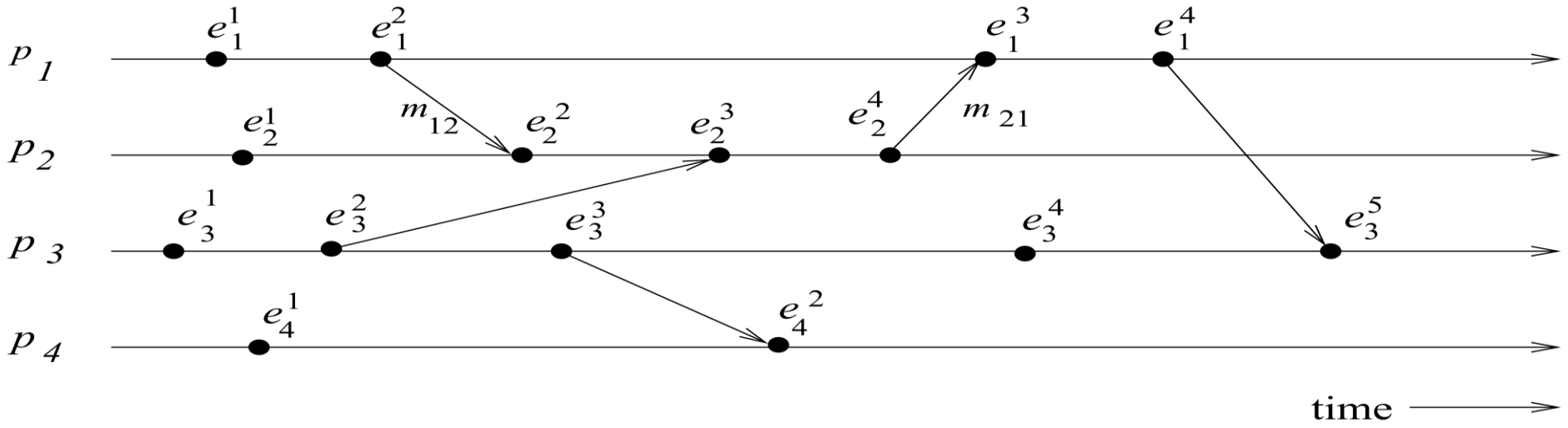
Example



- Which global states are consistent?

1) $\{LS^1_1, LS^3_2, LS^3_3, LS^2_4\}$
2) $\{LS^2_1, LS^4_2, LS^4_3, LS^2_4\}$
3) $\{LS^2_1, LS^1_2, SC^{12}_{12}, LS^3_3, LS^2_4\}$

Global States and Snapshots

- Recording the global state of a distributed system on-the-fly is an important paradigm.
- The lack of globally shared memory, global clock and unpredictable message delays in a distributed system make this problem non-trivial.
- Building on the definition consistent global states we discuss issues to be addressed to obtain consistent distributed snapshots.
- Then several algorithms to determine on-the-fly such snapshots are presented for different message delivery models.

Messages in Transit

- For a channel C_{ij} , the following set of messages can be defined based on the local states of the processes p_i and p_j

  Transit: $transit(LS_i, LS_j) = \{m_{ij} \mid send(m_{ij}) \in LS_i \wedge rec(m_{ij}) \notin LS_j \}$

Conditions for Consistent Global States

- The global state of a distributed system is a collection of the local states of the processes and the channels.
- Notation wise, a global state GS is defined as,
  - $GS = \{ \cup_i LS_i , \cup_{i,j} SC_{ij} \}$
- A global state GS is a consistent global state iff it satisfies the following two conditions :
- C1: Law of Conservation of Messages
- C2: For every effect, its cause must be present.

Conditions for Consistent Global States

- A global state GS is a consistent global state iff it satisfies the following two conditions :
- C1 states the law of conservation of messages.
  - Every message that is recorded as sent in the local state of some process is either captured in the state of the channel or is captured in the local state of the receiver.
  - $send(m_{ij}) \in LS_i \Rightarrow m_{ij} \in SC_{ij} \oplus rec(m_{ij}) \in LS_j$. ($\oplus$ is the Exclusive-OR operator)
- C2 states that for every effect, its cause must be present.
  - If a message is not recorded as sent in the local state of a process P_i, then the message cannot be included in the state of the channel C_{ij} or be captured as received by P_j.
  - $send(m_{ij}) \notin LS_i \Rightarrow m_{ij} \notin SC_{ij} \wedge rec(m_{ij}) \notin LS_j$.

Difficulty of Taking a Snapshot

- Issue 1: How to distinguish between the messages to be recorded in the snapshot from those not to be recorded.
  - Any message that is sent by a process before recording its snapshot, must be recorded in the global snapshot (from C1).
  - Any message that is sent by a process after recording its snapshot, must not be recorded in the global snapshot (from C2).
- Issue 2: How to determine the instant when a process takes its snapshot.
  - A process p_j must record its snapshot before processing a message m_{ij} that was sent by process p_i after recording its (p_i) snapshot.

Algorithms for Capturing a Snapshot

- Armed with our knowledge, we will now study algorithms for capturing a snapshot of a distributed system.
- We will see how the guarantees on message delivery helps simplify algorithms.
- We now consider systems where it is not necessary that every pair of processors share a direct channel between them.
- We will start with algorithms under the assumption that message delivery follows causal rules, then study algorithms for FIFO channels, and then for arbitrary channels.

Snapshots in a Causal Channels

- The causal message delivery property provides a built-in message synchronization to control and computation messages.
- Two global snapshot recording algorithms, namely, Acharya-Badrinath and Alagar-Venkatesan, exist that assume that the underlying system supports causal message delivery.

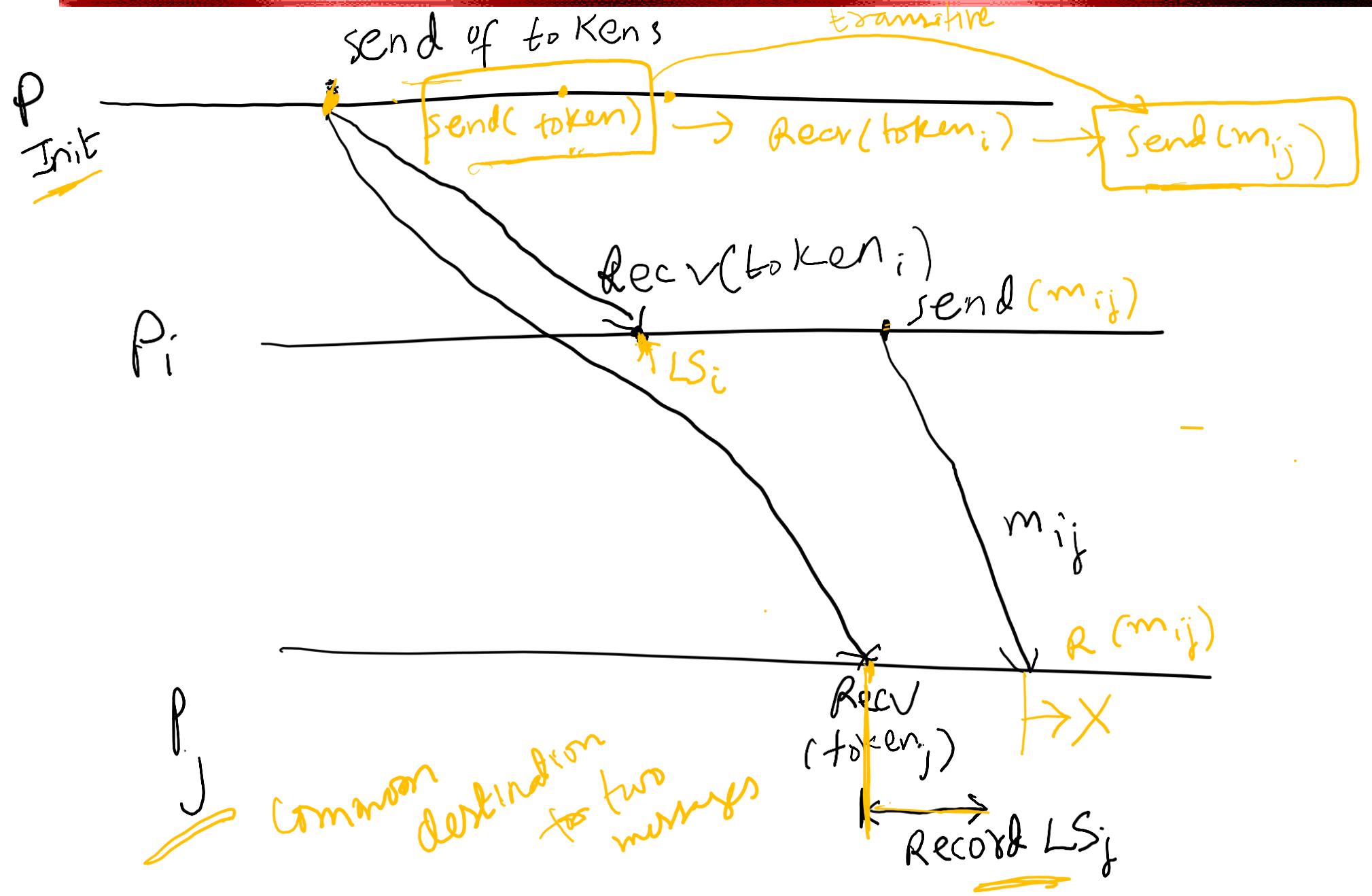
Snapshots in a Causal Channels

- In both these algorithms the recording of process state is identical and proceeds as follows :
- An initiator process broadcasts a token, denoted as `token`, to every process including itself.
- Let the copy of the token received by process p_i be denoted $token_i$.
- A process p_i records its local snapshot LS_i on receiving $token_i$ and sends the recorded snapshot to the initiator.
- The algorithm terminates when the initiator receives the snapshot recorded by each process.

Snapshots in a Causal Channels

- The correctness of the approach depends on the following observation.
- For any two processes p_i and p_j, the following property is satisfied:
  - $send(m_{ij}) \notin LS_i \Rightarrow rec(m_{ij}) \notin LS_j$.
- This is due to the causal ordering property of the underlying system as explained next.

Snapshots in a Causal Channels



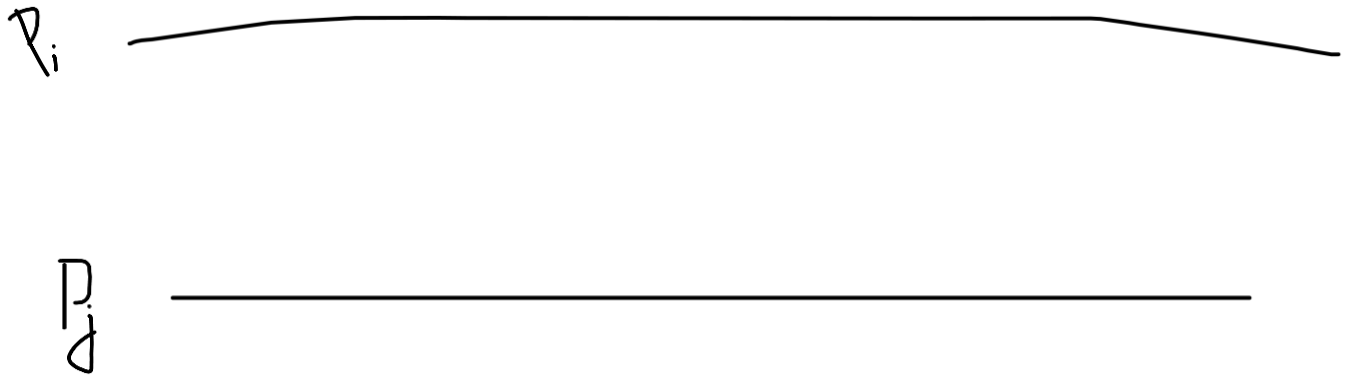
Snapshots in a Causal Channels

- Let a message m_{ij} be such that $rec(token_i) \rightarrow send(m_{ij})$
- Then $send(token_j) \rightarrow send(m_{ij})$ and the underlying causal ordering property ensures that $rec(token_j)$, at which instant process p_j records LS_j, happens before $rec(m_{ij})$.
- Thus, m_{ij}, whose send is not recorded in LS_i, is not recorded as received in LS_j.

Acharya – Badrinath Algorithm

- Each process p_i maintains arrays $SENT_i[1, ...N]$ and $RECD_i[1, ..., N]$.
- $SENT_i[j]$ is the number of messages sent by process p_i to process p_j.
- $RECD_i[j]$ is the number of messages received by process p_i from process p_j.
- Channel states are recorded as follows:
  - When a process p_i records its local snapshot LS_i on the receipt of token, it includes arrays $RECD_i$ and $SENT_i$ in its local state before sending the snapshot to the initiator.

Acharya – Badrinath Algorithm



- When the algorithm terminates, the initiator determines the state of channels as follows:
- The state of each channel from the initiator to each process is empty.
- The state of channel from process p_i to process p_j is the set of messages whose sequence numbers are given by $\{RECD_j[i] + 1, \ldots, SENT_i[j]\}$.

Acharya – Badrinath Algorithm

Complexity:

- This algorithm requires 2n messages and 2 time units for recording and assembling the snapshot, where one time unit is required for the delivery of a message.
- If the contents of messages in channels state are required, the algorithm requires 2n messages and 2 time units additionally.

Chandy-Lamport algorithm – FIFO Channels

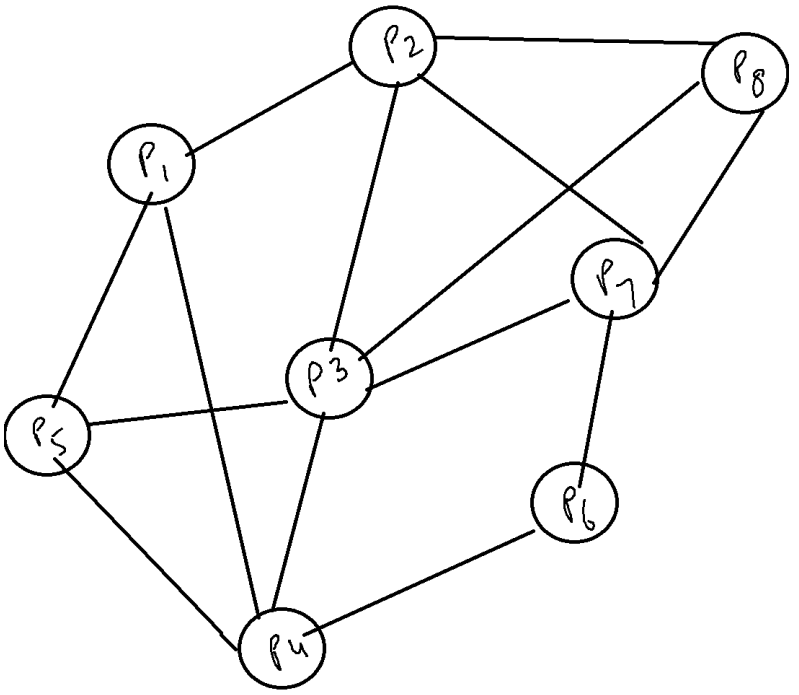
- This algorithm presented by Mani Chandy and Leslie Lamport works for FIFO channels.
- Uses the guarantees offered by the FIFO nature of message delivery of channels to solve Issue 1 and Issue 2.

Chandy-Lamport algorithm – FIFO Channels

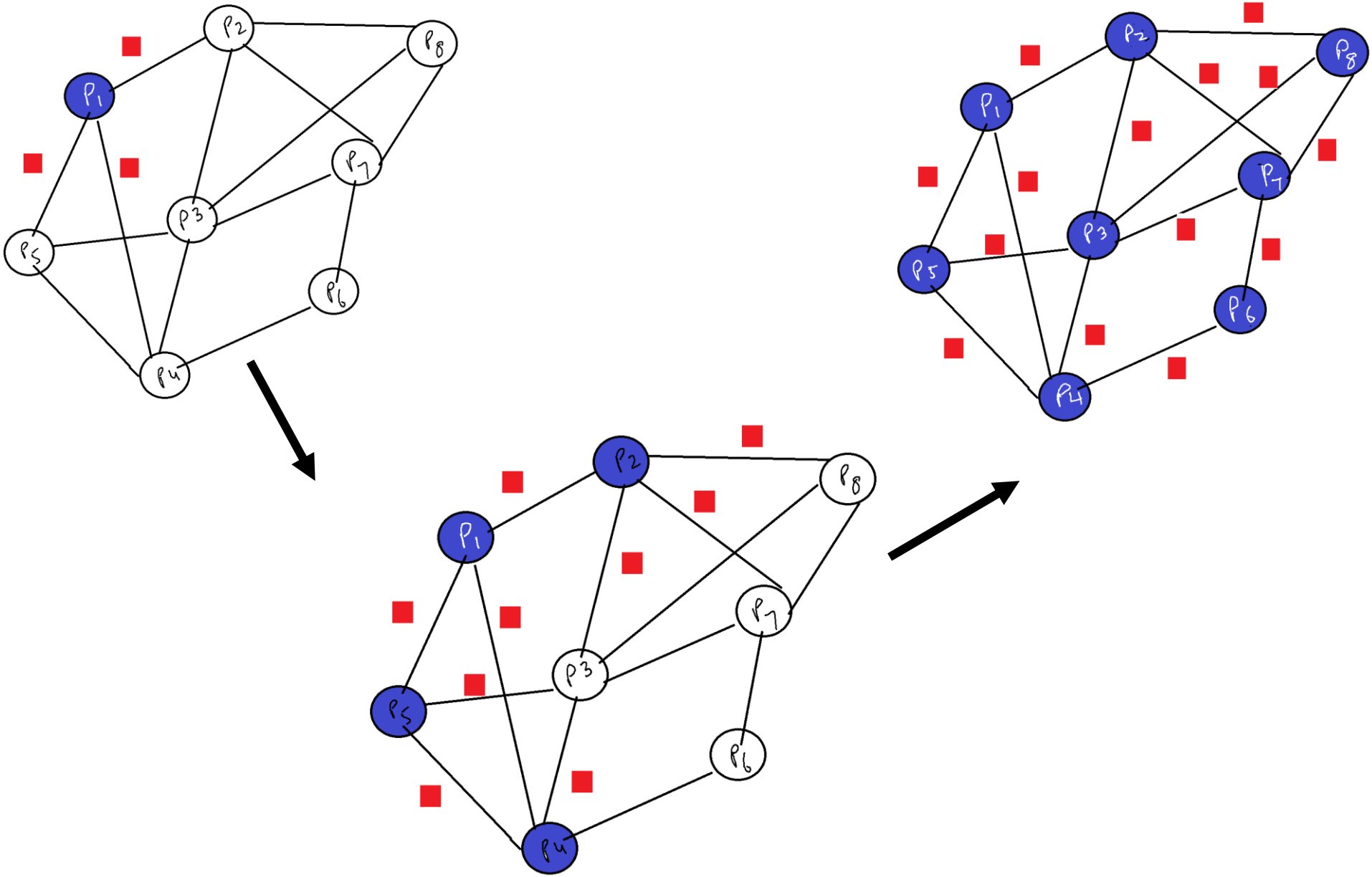
- The Chandy-Lamport algorithm uses a control message, called a ***marker*** whose role in a FIFO system is to separate messages in the channels.
- After a site has recorded its snapshot, it sends a marker, along all of its outgoing channels before sending out any more messages.
- This marker separates the messages in the channel into those to be included in the snapshot from those not to be recorded in the snapshot.
- A process must record its snapshot no later than when it receives a marker on any of its incoming channels.

Chandy-Lamport algorithm – FIFO Channels

- The algorithm can be initiated by any process by executing the "Marker Sending Rule" by which it records its local state and sends a marker on each outgoing channel.
- A process executes the "Marker Receiving Rule" on receiving a marker.
- If the process has not yet recorded its local state, it records the state of the channel on which the marker is received as empty and executes the "Marker Sending Rule" to record its local state.



Example Run



Chandy-Lamport algorithm – FIFO Channels

- The algorithm terminates after each process has received a marker on all of its incoming channels.
- All the local snapshots get disseminated to all other processes and all the processes can determine the global state.

Chandy-Lamport algorithm – FIFO Channels

- Marker Sending Rule for process i

1. Process i records its state.
2. For each outgoing channel C on which a marker has not been sent, Process i sends a marker along C before i sends further messages along C.

- Marker Receiving Rule for Process j

On receiving a marker along channel C:

- if Process j has not recorded its state then
  - Record the state of C as the empty set
  - Follow the "Marker Sending Rule"
- else
  - Record the state of C as the set of messages received along C after j's state was recorded and before j received the marker along C

Correctness and Complexity

- Correctness
- Due to FIFO property of channels, it follows that no message sent after the marker on that channel is recorded in the channel state. Thus, condition C2 is satisfied.
- When a process p_j receives message m_{ij} that precedes the marker on channel C_{ij}, it acts as follows: If process p_j has not taken its snapshot yet, then it includes m_{ij} in its recorded snapshot. Otherwise, it records m_{ij} in the state of the channel C_{ij}. Thus, condition C1 is satisfied.

Correctness and Complexity

- Complexity
  - The recording part of a single instance of the algorithm requires O(e) messages and O(d) time, where e is the number of edges in the network and d is the diameter of the network.